

Internet of Things

By

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Course Materials

► Textbook



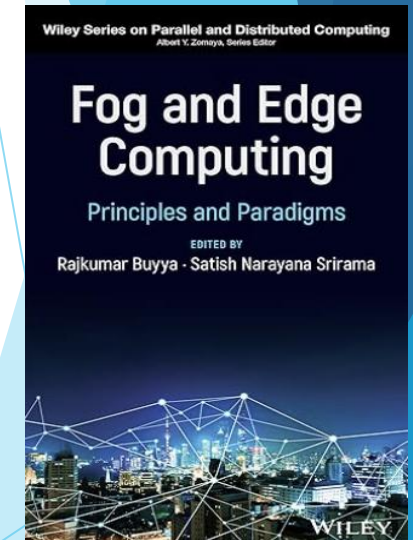
IOT Architecture and design principles



Designing the IOT



AI for IOT



Fog and Edge computing (Architecture)

Reference materials will be provided

Course overview

- ▶ Course Evaluation:
 - ▶ Mid-term Exam 30%
 - ▶ Final Exam 30%
 - ▶ Quiz 10%
 - ▶ Homework-Assignment 15%
 - ▶ Attendance + class activity 5%
- ▶ Small project + paper writing 10%

Course Syllabus

Week	Content
Week one	Introduction of IoT
Week two	Application and Evolution of IoT and its concepts
Week three	Design principals
Week four	Design principals and Architecture
Week five	Platforms and Arduino and Raspberry pi
Week six	Connectivity technologies
Week seven	Sensor Network
Week eight	Mid term exam

Course Syllabus

Week	Content
Week nine	Arduino and Raspberry pi prototyping process
Week Ten	Cloud and Fog computing for IoT
Week Eleven	IoT privacy and security issue
Week Twelve	Cryptography & Hashing
Week Thirteen	Blockchain Basics and Consensus
Week Fourteen	Blockchain architecture
Week Fifteen	Middle Ware, Analytics and Application
Week Sixteen	Final Exam

Outline

- ▶ Introduction of IoT
- ▶ Features of IoT
- ▶ Technology used
- ▶ Framework
- ▶ Applications
- ▶ Issues and Challenges
- ▶ Conclusion
- ▶ References

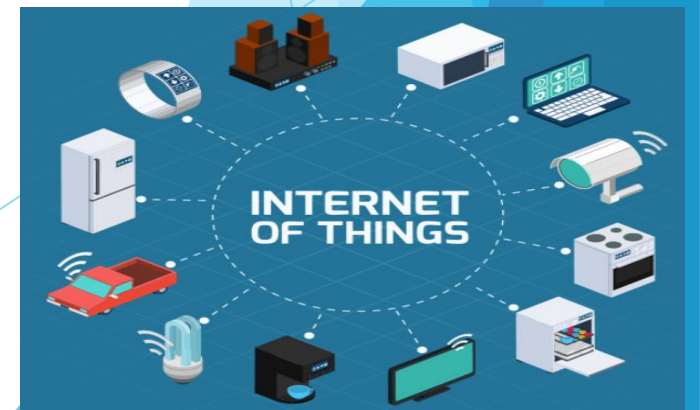
Introduction to IoT

IoT system



Introduction of IoT

- ▶ The Internet of Things is the network of physical objects—devices, vehicles, buildings and other items embedded with electronics, software, sensors, and network connectivity—that enables these objects to collect and exchange data.
- ▶ IoT is connecting every physical object in the world using wireless.
- ▶ Internet of Things are able to collect and exchange data using embedded sensors.



Introduction to IoT

- ▶ The concept of IoT began with things classified as identity communication devices. Radio Frequency Identification Device (RFID) is an example of an identity communication device. Things are tagged to these devices for their identification in future and can be tracked, controlled and monitored using remote computers connected through the internet.
- ▶ The concept of IoT enables, for example GPS based tracking, controlling and monitoring of devices; machine to machine (M2M) communication; connected cars; communication between wearable and personal devices and Industry 4.0.
- ▶ The IoT concept has made smart cities a reality and is also expected to make self-driving cars functional very soon.

IoT: Growth

Year	Number of Connected Devices
1990	0.3 Million
2000	90 Million
2010	5 Billion
2015	9 Billion
2025	1 trillion (Expected)



IoT: Technologies

- ▶ RFID
- ▶ WiFi
- ▶ Barcode
- ▶ QR Code
- ▶ ZigBee
- ▶ Sensors and Smartphones

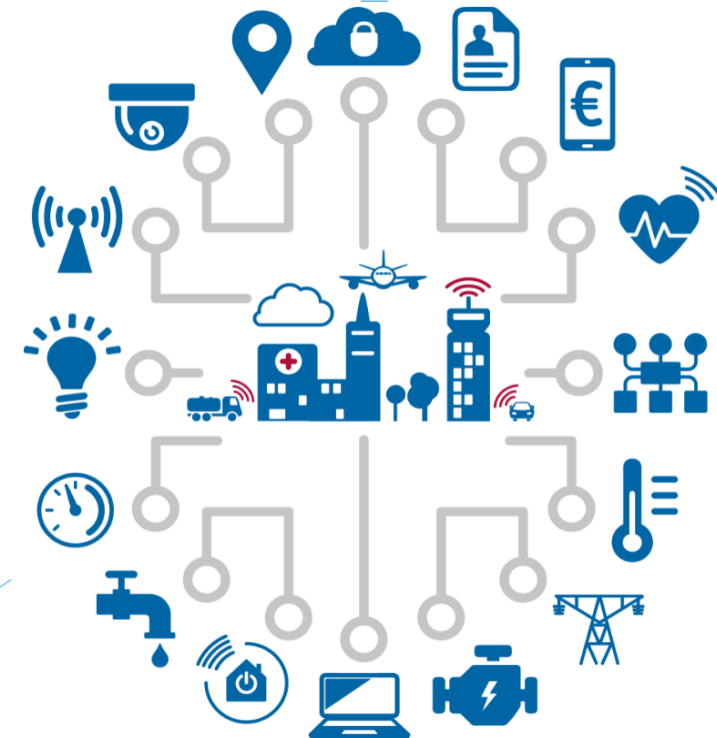


IoT: Characteristics Features, Advantages, Disadvantages

Characteristics	Features	Advantages	Disadvantages
Intelligence	Device Virtualization	Integration of Devices	Security
Connectivity	High Speed	Reliable, Secure Bi-Directional Communication	Privacy
Sensing	End Point Management	Enhancement Data Collection	Complexity
Expressing	Small Devices	Reduce Waste	Flexibility
Energy	Stream Processing	Real Time Analysis Improved Engagement	Compliance
Safety	Data Enrichment	Automation and Technology Optimization	Tier Management

IoT Vision

- ▶ IoT is a vision where things (wearable watches, alarm clocks, home devices , surrounding objects) become smart and function like a living entities by sensing, computing and communicating through embedded devices which interact with remote objects (servers, clouds, applications, services and processes) or persons through the Internet or Near-Field Communication (NFC) etc.
- ▶ The vision of IoT can be understood through Examples 1.1 and 1.2.



Internet of Things concept

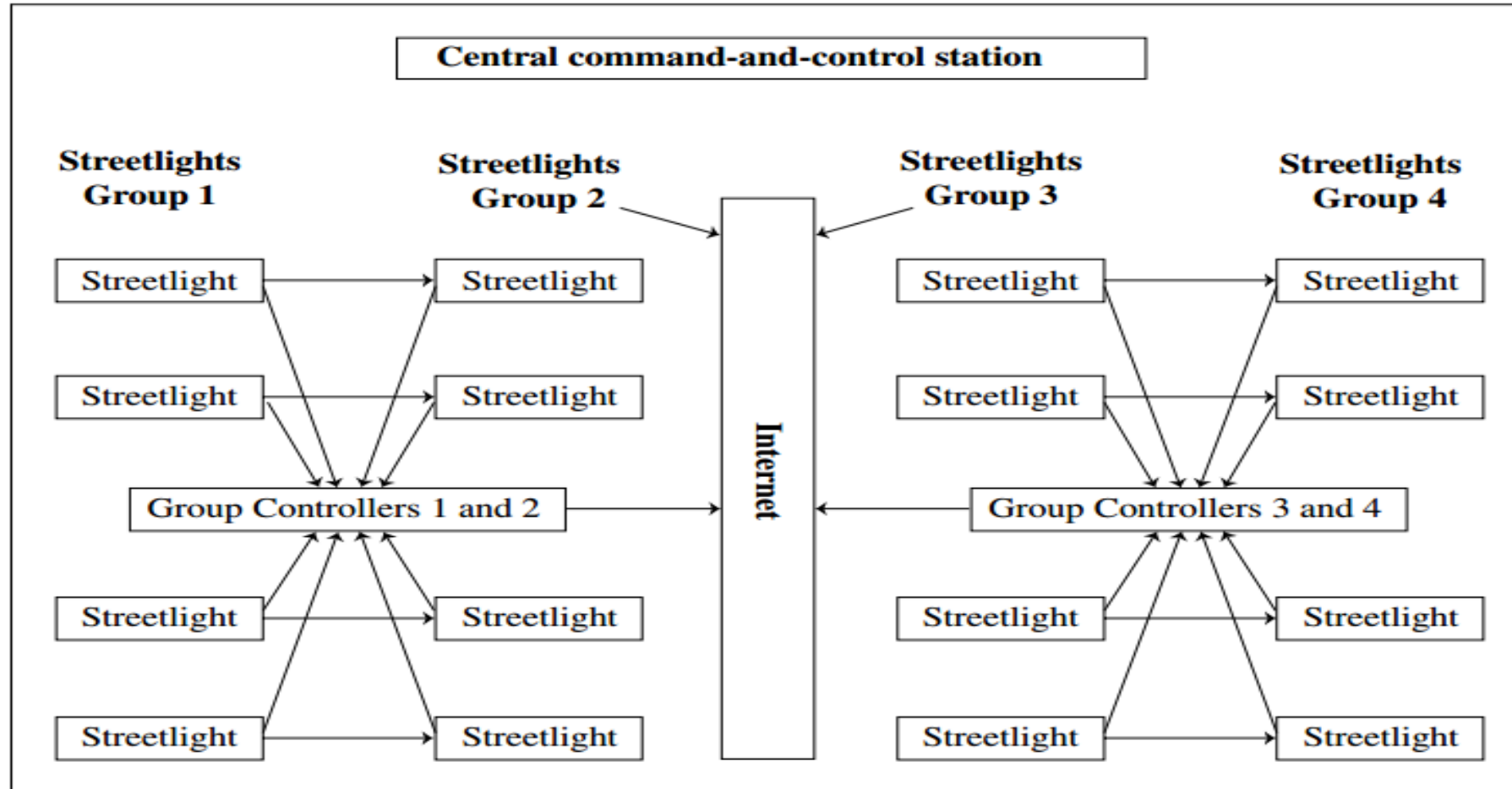


Figure 1.1 Use of Internet of Things concept for street lights in a city

Internet of Things: Architecture and Design principles

Street lights in a city can be made to function like living entities through sensing and computing using tiny embedded devices that communicate and interact with a central control-and-command station through the Internet.

Assume that each light in a group of 32 streetlights comprises a sensing, computing and communication circuit.

Each group connects to a group-controller (or coordinator) through Bluetooth or ZigBee.

Each controller further connects to the central command-and-control station through the Internet.

Internet of Things: Architecture and Design principles

The station receives information about each street light in each group in the city at periodic intervals. The information received is related to the functioning of the 32 lights, the faulty lights, about the presence or absence of traffic in group vicinity, and about the ambient conditions, whether cloudy, dark or normal daylight.

Internet of Things: Architecture and Design principles

The station remotely programs the group controllers, which automatically take an appropriate action as per the conditions of traffic and light levels. It also directs remedial actions in case a fault develops in a light at a specific location. Thus, each group in the city is controlled by the 'Internet of street lights'.

Figure 1.1 shows the use of the IoT concept for streetlights in a city.

Internet of Things

As per Collins Dictionary, hyperconnectivity means use of multiple systems and devices to remain constantly connected to social networks and streams of information.

- ▶ Physical Object + Controller, Sensor and Actuators + Internet = IoT (e.g. umbrella)
- ▶ Gather + Enrich + Stream + Manage + Acquire + Organize and Analyze = IoT with connectivity to Data Centre, Enterprise or Cloud Server (e.g. street light)
- ▶ Gather + Consolidate + Connect + Collect + Assemble + Manage and Analyze = IoT (e.g. IBM conceptual framework figure 1.3)

Internet of Things

- ▶ Equation 1.3 represents a complex conceptual framework for IoT using cloud-platform based processes and services. The steps are as follows:
- ▶ Levels 1 and 2 consist of a sensor network to gather and consolidate the data.
- ▶ First level gathers the data of the things (devices) using sensors circuits. The sensor connects to a gateway.

A general framework for IoT

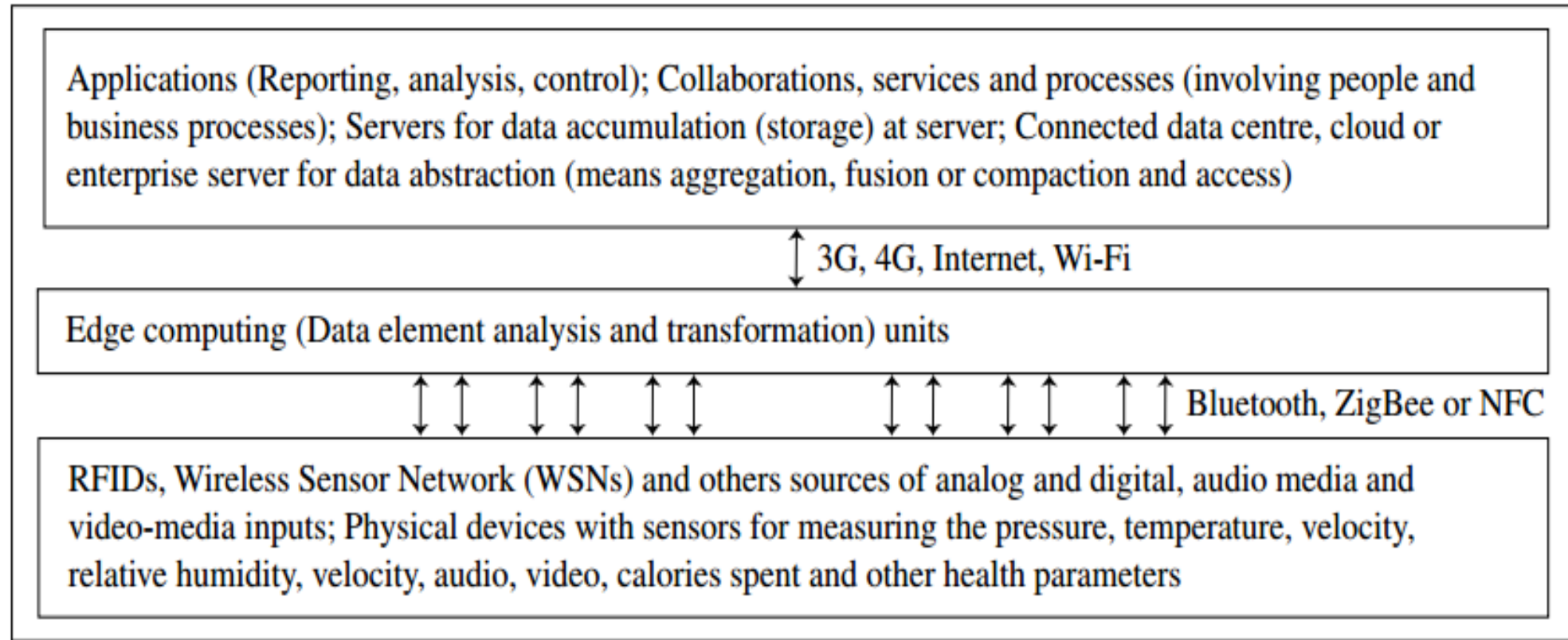


Figure 1.2 A general framework for IoT using smart and hyperconnected devices, edge computing and applications

Internet of Things: Architecture and Design principles

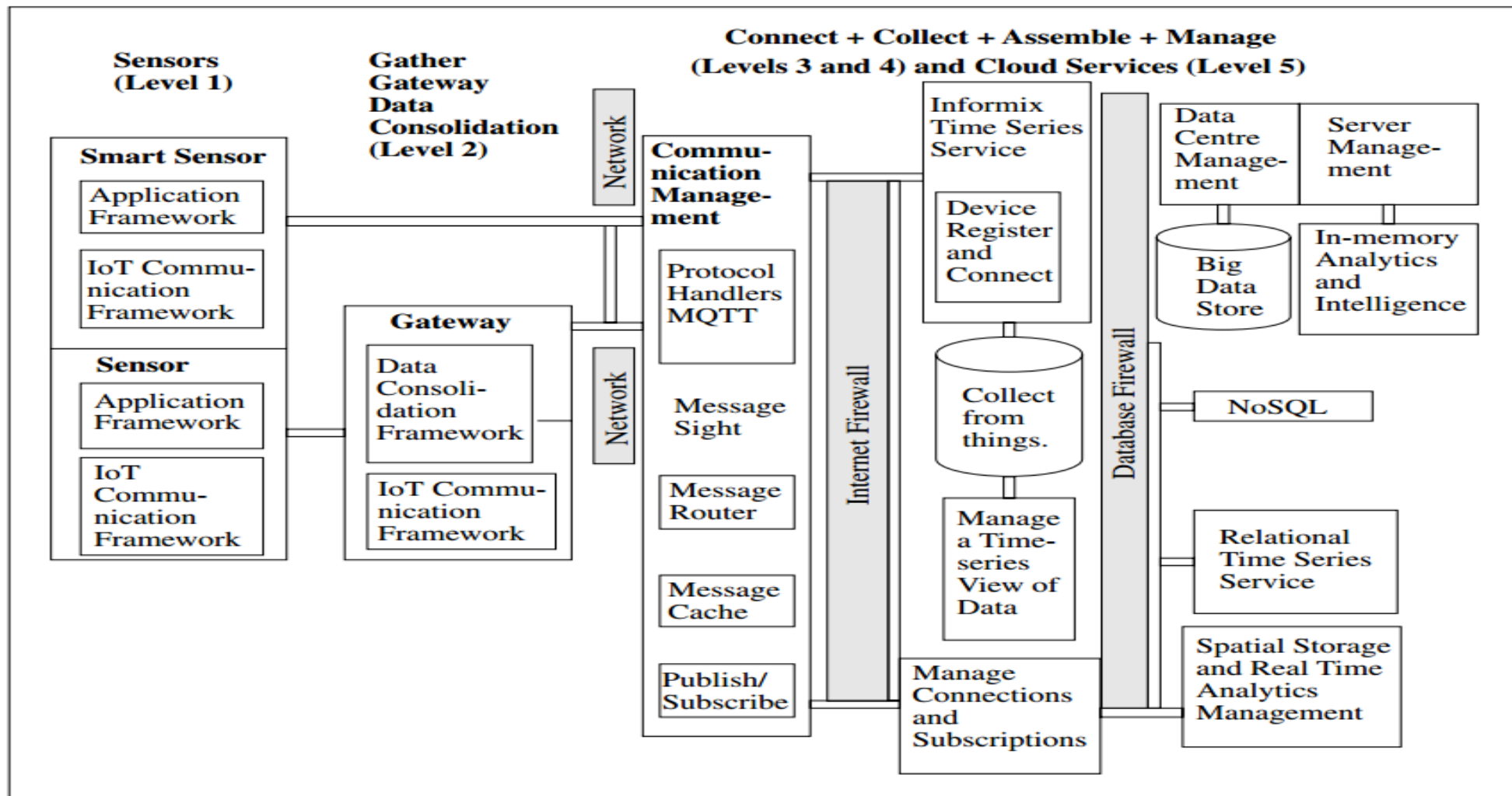


Figure1.3 IBM IoT conceptual framework

Internet of Things: Architecture and Design principles

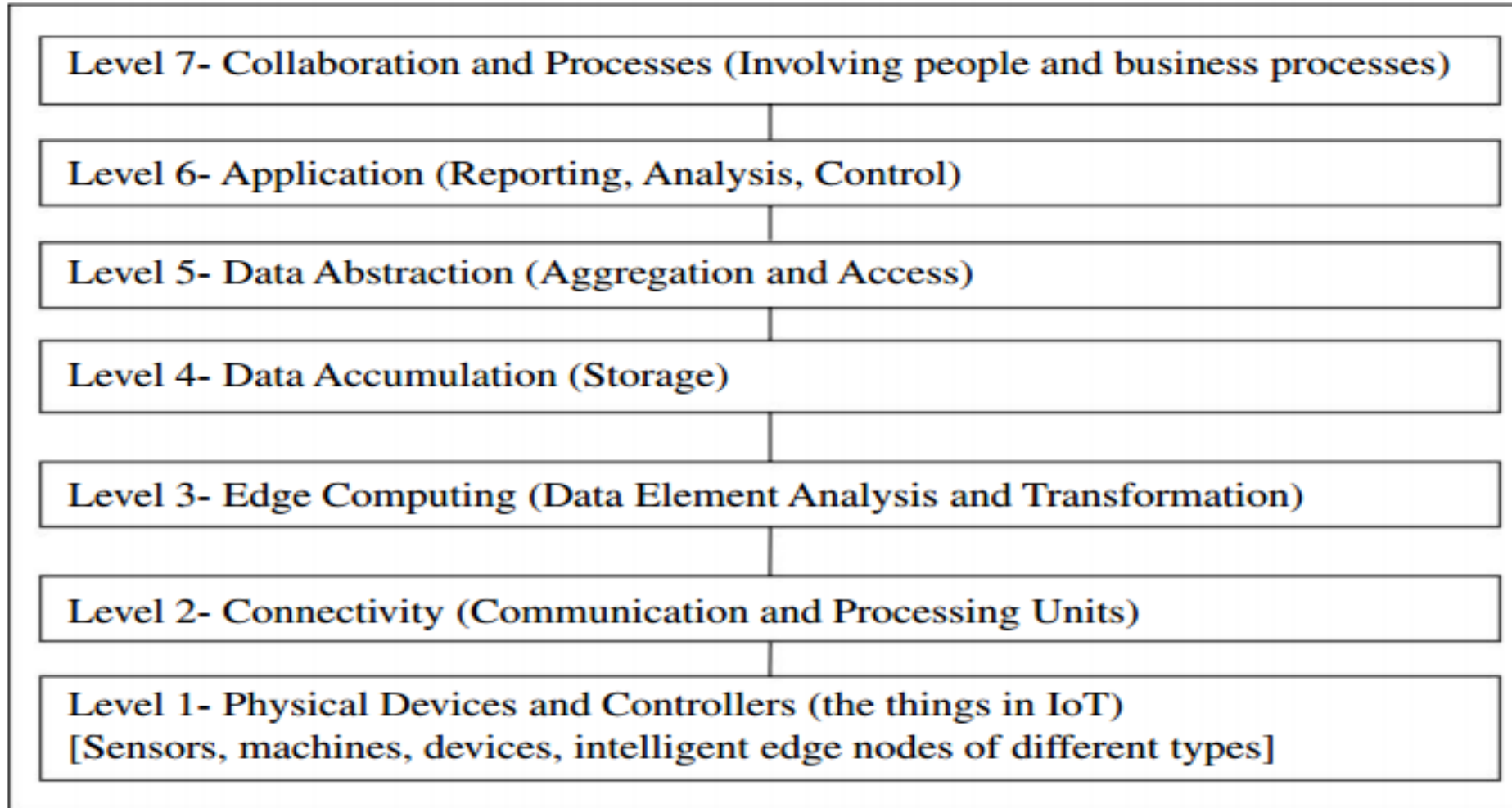
IBM IoT conceptual framework blocks and components are the basis of this equation.

- ▶ In general, things refer to an internetwork of devices and physical objects. This framework consists of a number of subsystems. The data is acquired at remote locations in a database or data store.

The services and processes need data managing, acquiring, organising and analysing.

- ▶ Various conceptual frameworks of IoT find number of applications including the ones in M2M communication networks, wearable devices, city lighting, security and surveillance and home automation. Smart systems use the things (nodes) which consist of smart devices, smart objects and smart services. Smart systems use the user interfaces (UIs), application programming interfaces (APIs), identification data, sensor data and communication ports.

IoT: Architectural View



CISCO seven leveled reference model

Figure 1.4 An IoT reference model suggested by CISCO that gives a conceptual framework for a general IoT system

IoT: Architectural View

An IoT system has multiple levels (Equations 1.1 to 1.3). These levels are also known as tiers. A model enables conceptualization of a framework. A reference model can be used to depict building blocks, successive interactions and integration. An example is CISCO's presentation of a reference model comprising seven levels (Figure 1.4).

Oracle suggested IoT architecture

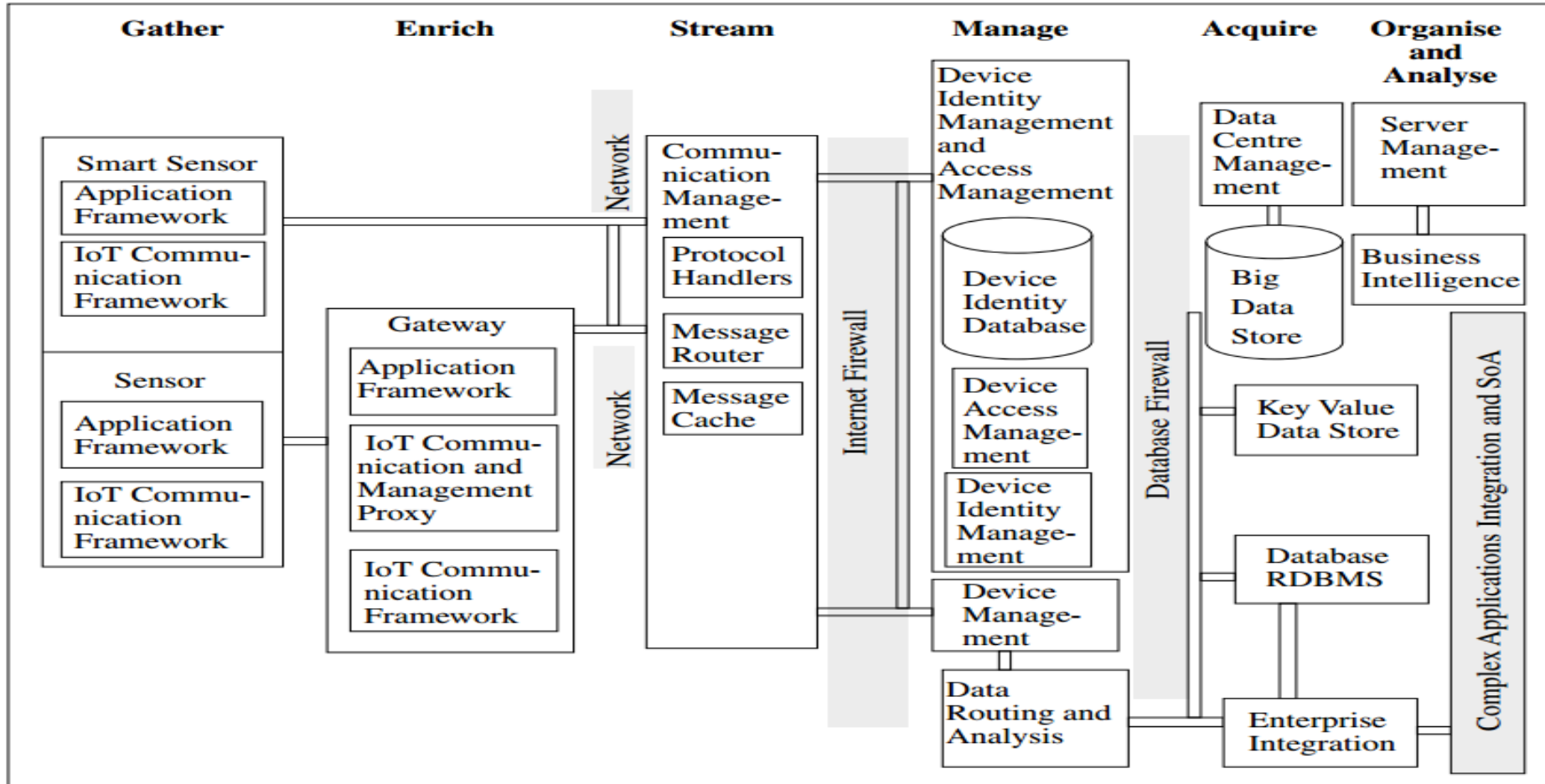


Figure 1.5 Oracle's IoT architecture (Device identity management means identifying a device, registering a device for actions after identifying, de-registering the device, assigning unique identity to the device. Device access management means enabling, disabling the device access, authenticating a device for access, authorizing a device for access to a subsystem.

Oracle suggested IoT architecture

An architecture has the following features:

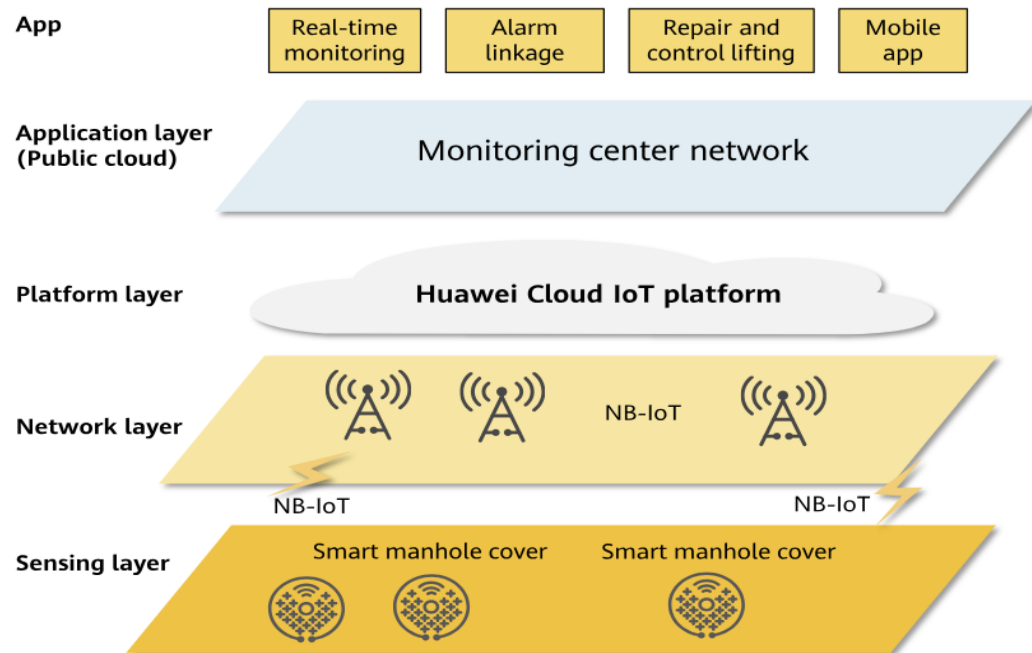
- ▶ The architecture serves as a reference in applications of IoT in services and business processes.
- ▶ A set of sensors which are smart, capture the data, perform necessary data element analysis and transformation as per device application framework and connect directly to a communication manager.
- ▶ A set of sensor circuits is connected to a gateway possessing separate data capturing, gathering, computing and communication capabilities. The gateway receives the data in one form at one end and sends it in another form to the other end.

Oracle suggested IoT architecture

- ▶ The communication-management subsystem consists of protocol handlers, message routers and message cache.
- ▶ This management subsystem has functionalities for device identity database, device identity management and access management.
- ▶ Data routes from the gateway through the Internet and data center to the application server or enterprise server which acquires that data.

Case study

Smart Manhole Cover Solution



Customer benefits

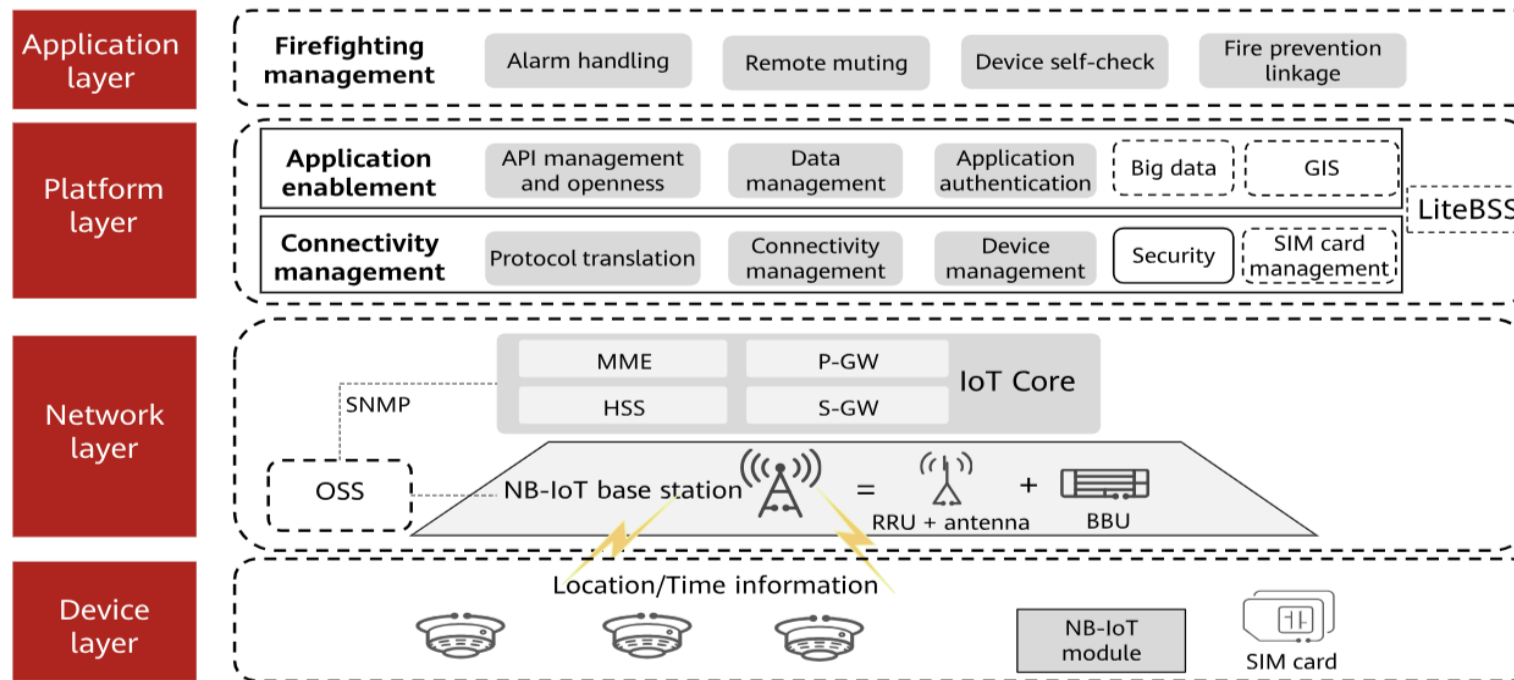
- **Construction of shared collection devices in cities:** Unified installation of functional facilities in multiple cities, and unified data aggregation and backhaul reduce the construction costs of city infrastructure.
- **Platform-based and unified O&M:** Unified O&M and monitoring of sensors mounted in each bureau improves O&M efficiency and reduces costs.
- **Alarm reporting and anti-theft:** By monitoring manhole covers in real time, the system can detect incidents (theft, displacement, and damage), generate alarms, as well as notify construction organizations or policing platforms to take immediate action, eliminating security risks and ensuring city security.

Scenarios

- **Real-time monitoring:** Manhole covers in a large area are monitored in real time and intelligently maintained.
- **Alarm reporting:** Monitoring manhole covers that are abnormally open helps identify incidents such as theft, displacement, and damage. Alarm reports send the location to the monitoring center and policing platform. Then the monitoring center schedules construction vehicles to maintain these manhole covers and the policing platform dispatches officers to the incident location.

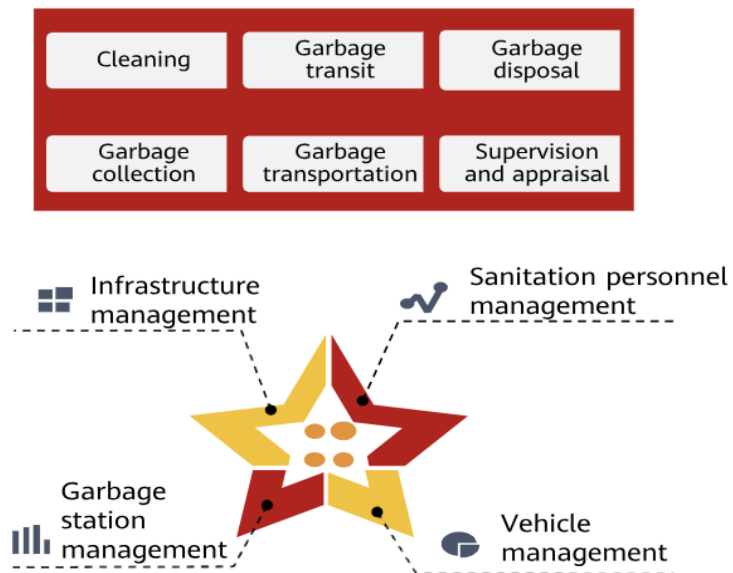
Case study

Smart Firefighting Solution



Case study

Smart Sanitation Solution



Manages people, vehicles, objects, and events in real time.

1. Install RFID tags

2. Read tags when lifting garbage bins

4G, 5G, etc.

Control center

Vehicle track and real-time location

Garbage clearance statistics in the cloud



Case study :smart sanitation

- Manages the entire process of sanitation personnel, vehicles, and garbage cleaning in real time.
- Supervises the operating time, routes, fuel consumption, and water-spraying of sanitation vehicles.
- Monitors garbage bin status, garbage collection routes, and garbage transit stations in real time.
- The solution is based on real-time, visualized video surveillance. It enables refined management, standardizes operations, and intelligently appraises urban sanitation, making sanitation orderly, effective, and controllable.

Major Components of IoT System

1. **Physical object** (with embedded software into a hardware)
2. **Hardware** (consisting of a microcontroller, firmware, sensors, control unit, actuators and communication module)
3. **Communication Module** (Software consisting of device APIs and device interface for communication over the network and communication circuit/port(s), and middleware for creating communication stacks using 6LowPAN, CoAP, LWM2M, IPv4, IPv6 and other protocols.)
4. **Software** (for actions on messages, information and commands which the devices receive and then output to the actuators, which enable actions such as glowing LEDs, robotic hand movement etc.)

Major Components of IoT System

► Sensor and Control Units

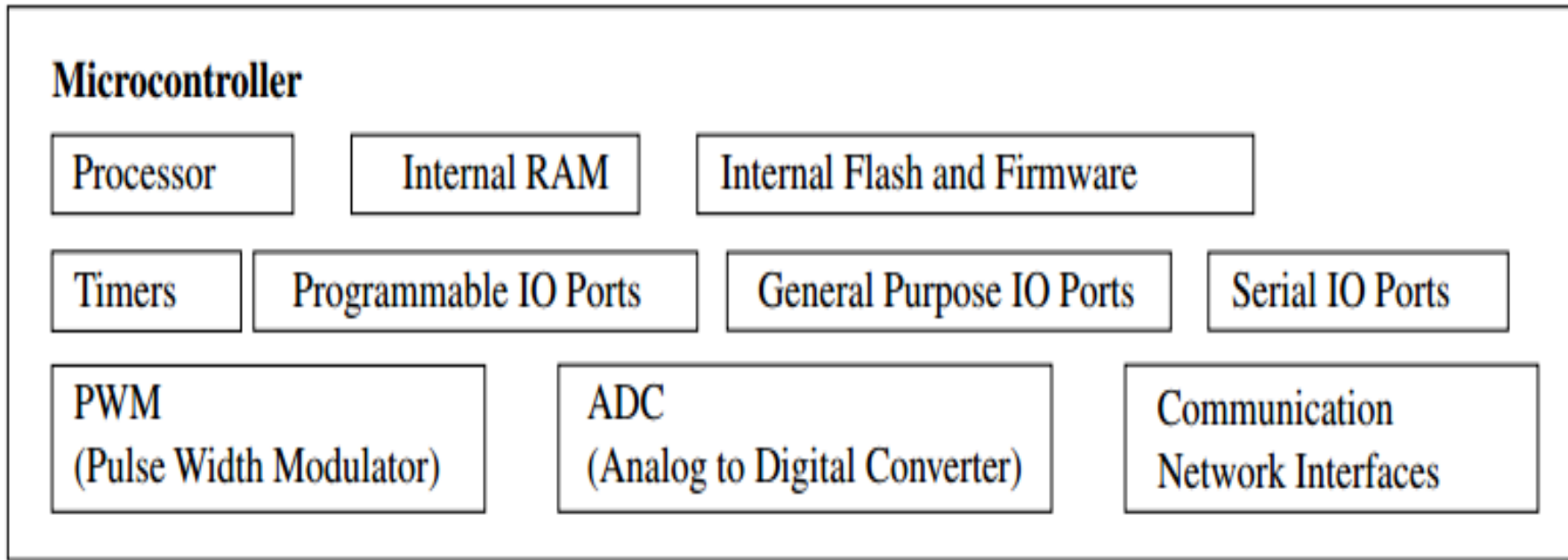


Figure 1.6 Various functional units in an MCU that are embedded in an IoT device or a physical object

Major Components of IoT System

An MCU comprises a processor, memory and several other hardware units which are interfaced together. It also has firmware, timers, interrupt controllers and functional IO units. Additionally, an MCU has application-specific functional circuits designed as per the specific version of a given microcontroller family. For example, it may possess Analog to Digital Converters (ADC) and Pulse Width Modulators (PWM).

Figure 1.6 shows various functional units in an MCU that are embedded in an IoT device or a physical object.

Communication Module

A communication module consists of protocol handlers, message queue and message cache. A device message-queue inserts the messages in the queue and deletes the messages from the queue in a first-in first-out manner. A device message-cache stores the received messages.

Representational State Transfer (REST) architectural style can be used for HTTP access by GET, POST, PUT and DELETE methods for resources and building web services.

IOT: Hardware, Middleware and Software

Hardware	Middleware	Software
Google Cloud IoT	Oracle-Fusion	Arduino
Arduino Nano Pro Mini	OpenIoT	Eclipse IoT
Raspberry-Pi	OpenRemote	Google Brillo
IBM Watson	OpenHUB	IBM IoT Foundation
Azure	Kaa	Azure IoT Suit
Amazon AWS IoT Core		Cloud Sensor
INTEL JOULE		Ninja Sphere
Netduino		Control Any
Flutter		



arduino



raspberry pi

Popular IoT Development Boards

▶ Arduino Yún

Arduino Yún board uses microcontroller ATmega32u4 that supports Arduino and includes Wi-Fi, Ethernet, USB port, micro-SD card slot and three reset buttons. The board also combines with Atheros AR9331 that runs Linux.

▶ Microduino

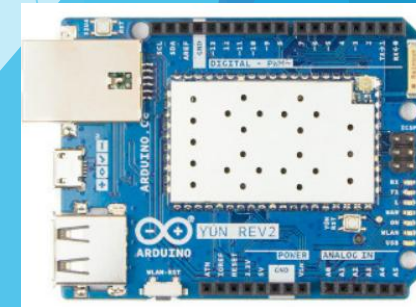
Microduino is a small board compatible with Arduino that can be stacked with the other boards. All the hardware designs are open source.

▶ Intel Galileo

Intel Galileo is a line of Arduino-certified development boards. Galileo is based on Intel x86 architecture. It is open-source hardware that features the Intel SOC X1000 Quark based Soc.



Raspberry Pi



Arduino Yun

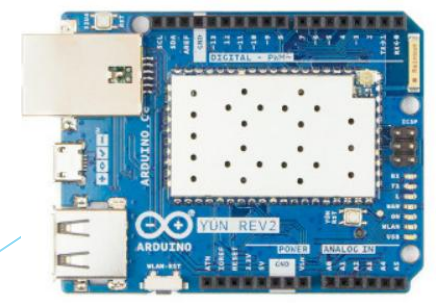
Popular IoT Development Boards

- ▶ Raspberry Pi Wireless Inventors Kit (RasWIK)

RasWIK (Raspberry Pi Wireless Inventors Kit) enables Raspberry Pi Wi-Fi connected devices. It includes documentation for 29 different projects or you can come up with one of your own.



Raspberry Pi



Arduino Yun

IoT software components for device hardware

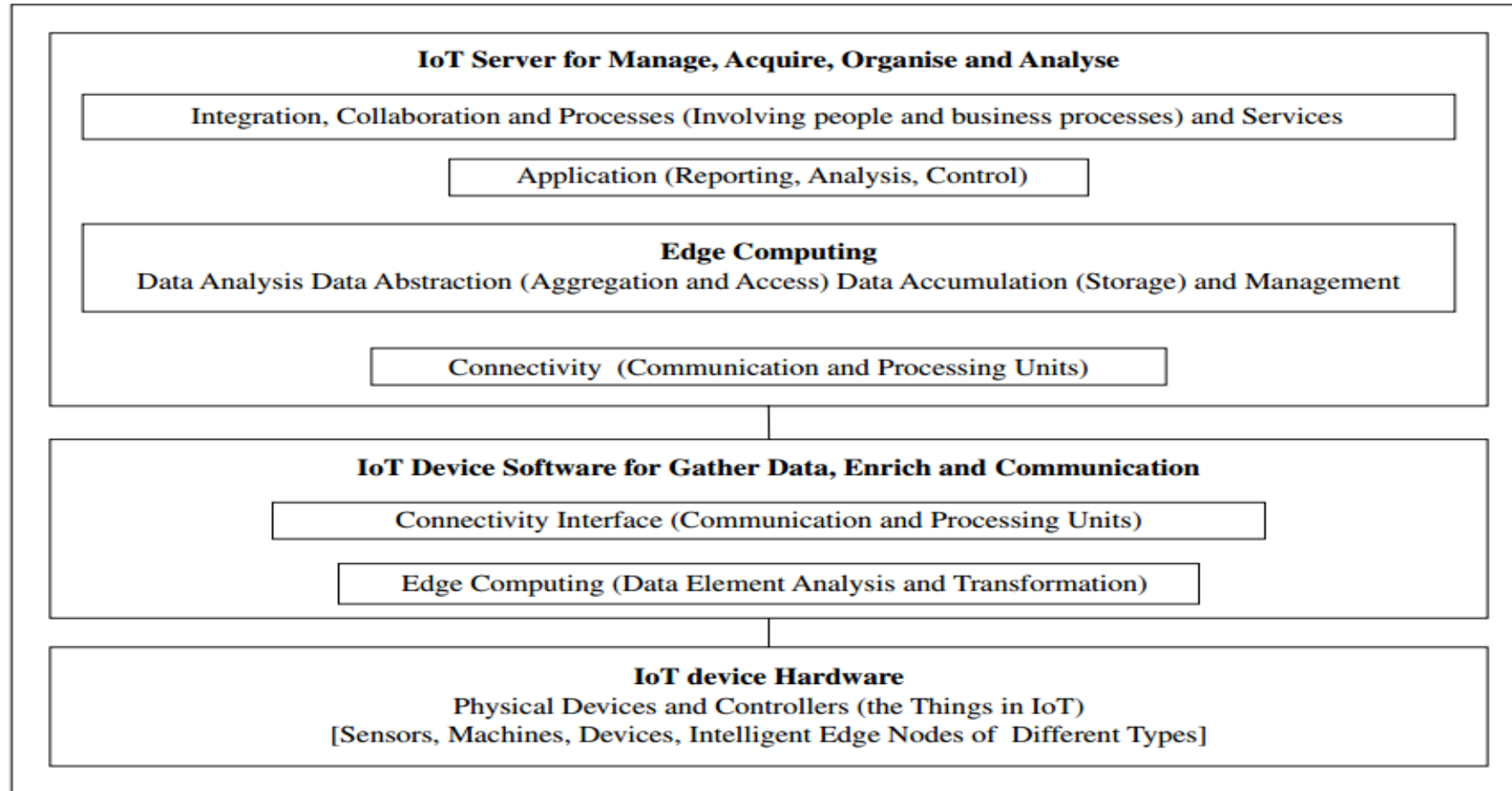


Figure 1.7 IoT software components for device hardware

IoT software components for device hardware

- ▶ IoT software consists of two components—software at the IoT device and software at the IoT server.
- ▶ OpenIoT is an open source middleware. It enables communication with sensor clouds as well as cloud-based ‘sensing as a service’. IoTsyS is a middleware which enables provisioning of communication stack for smart devices using IPv6, oBIX, 6LoWPAN, CoAP and multiple standards and protocols. The oBIX is standard XML and web services protocol oBIX (Open Building Information Xchange)

IoT software components for device hardware

► Firmware

Things Mist is an open-source firmware for true Internet-connectivity to the IoT. It enables resilient wireless mesh networking. Several microcontrollers with a range of wireless radios support Things MIST.

Eclipse IoT provides open-source implementation of standards such as MQTT CoAP, OMA-DM and OMA LWM2M, and tools for working with Lua, services and frameworks that enable an Open Internet of Things.

IoT software components for device hardware

Arduino development tools provide a set of software that includes an IDE and the Arduino programming language for a hardware specification for interactive electronics that can sense and control more of the physical world.

Kinoma S/W platform: Kinoma Create (kit for prototyping), Kinoma Studio development environment and Kinoma Platform Runtime are three different open-source projects. Kinoma Connect is a free app for iOS and Android smartphones and tablets with IoT devices.

Platforms and Integration Tools

ThingSpeak is an open data platform with an open API. It consists of APIs that enable real-time data collection, geolocation data, data processing and visualizations. It enables device status messages and plugins. It can process HTTP requests and store and process data. It can integrate multiple hardware and software platforms. It supports Arduino, Raspberry Pi, ioBridge/RealTime.io, and Electric Imp. An important feature of ThingSpeak is the support to MATLAB data analytics, mobile, web applications and social networks.

Nimbits is a cloud platform which supports multiple programming languages, including Arduino, JavaScript, HTML or Java library.

Platforms and Integration Tools

The software deploys on Google App Engine, any J2EE server on Amazon EC2 or Raspberry Pi. It processes a specific type of data and can also store the data. The data can be time- or geo-stamped.

IoT Toolkit offers Smart Object API, HTTP-to-CoAP Semantic mapping and a variety of tools for integrating multiple IoT-related sensor networks and protocols.

IoT: Application

- ▶ Traffic Monitoring
- ▶ Healthcare
- ▶ Agriculture
- ▶ Security
- ▶ Transport and Logistics
- ▶ Connected Cars
- ▶ Smart City
- ▶ Smart Home
- ▶ Wearables Devices



IoT: Issues and Challenges

- ▶ Privacy
- ▶ Security
- ▶ Connectivity



IoT: Future scope

- ▶ Wireless Sensor Networks
- ▶ Cloud Computing
- ▶ Big Data
- ▶ Distributed systems
- ▶ Mobile Computing
- ▶ Artificial Intelligence
- ▶ Machine Learning
- ▶ Security
- ▶ Semantic Web
- ▶ RFID
- ▶ Smart Cities

IoT: Future scope

- ▶ IoT: Future scope
- ▶ Hybrid AI
- ▶ Smart IoT
- ▶ IoT in Agriculture and farming
- ▶ Health
- ▶ Tracking and Shipping of goods
- ▶ Daily life



M2M Architecture

Machine-to-machine (M2M) refers to the process of communication of a physical object or device at machine with others of the same type, mostly for monitoring but also for control purposes.

IoT technology in industry involves the integration of complex physical machinery M2M communication with the networks of sensors, and uses analytics, machine learning, and knowledge discovery software.

References

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Question



Thank you

